

CSC-335

HCI & Computer Graphics

Lec 10: From Wireframe to Prototype

What is a Prototype?

- A representation of a design made before the final solution exists

Why Prototype?

- You often don't know how exactly the system should work
- Engineering and software development are expensive and time-consuming
- Maximizes the number of times you get to revise and refine your design before committing to code
- Each prototype is intended to answer one or more questions to help designers make decisions needed to advance their design.
- Prototyping enables testing of and receiving feedback on
 - Overall design concept
 - Functionality of different components of a system
 - User interactions
 - Layouts
 - Fine-grain design details like fonts and color schemes

Types of Prototypes

- Physical Prototypes
 - Function
 - Form
- Service Prototypes
 - Person-to-Person
 - Person-to-Technology
- Environment Prototypes



Wireframes

- A visual representation screens of an interactive application that shows layout, types of information that are displayed, and elements of pointer-based navigation.



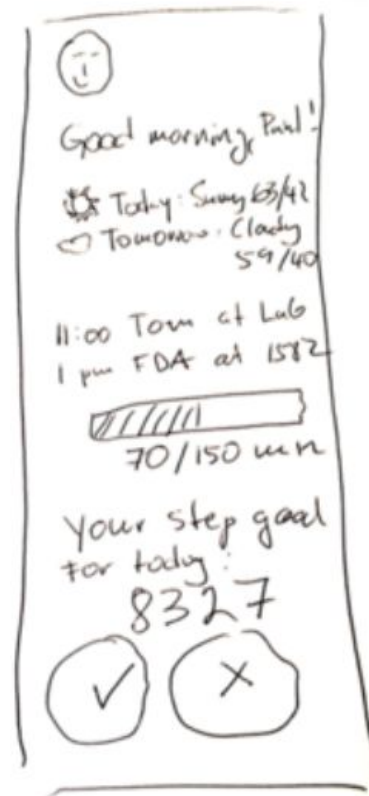
Questions Wireframes can Answer

- Do screens capture right chunks of system functionality?
- Are the components of a wireframe the right things to have on a single screen?
- Does the screen capture the right way those components should be presented to the user?
- Does the overall layout of components make sense?
- Do screens provide the right navigational elements?
- Would it be useful for users to receive a message with their progress and information about the day every morning?
- Is this the right information to present?
- Is the order of the information correct?
- Are these the right response options for step goal?
- How would user finish interacting with this message?

Questions Wireframes Do NOT Answer!

- How should content be ordered? (e.g., results in a list)
- How does the user transition among multiple screens? (navigation often underspecified)
- What is the right visual design for the screen?
- What non-visual output is the system producing?

How to wireframe?



When to Create Wireframes?

- When the designer understands a chunk of functionality and wants to get feedback
- When there are several ways something can be presented to users, and the designer needs to choose among them
- When developers need to start planning system backend

	Wireframe	Low-Fi Prototype
Fidelity	Very low	Low
Interactivity	None	Basic (clickable)
Purpose	Layout & structure	Flow & usability testing
Output	Static document	Clickable model
Tools	Figma, Balsamiq, pen & paper	Figma, Marvel, InVision

Low-Fidelity Interactive Prototypes

- A representation of a design, made with lightweight tools and materials, that enables early testing of concepts, layouts, and user interactions.
- Lo-fi prototyping maximizes the number of times designers get to revise and refine their design before it is committed to code.

How to Prototype

- Start with wireframes for each screen of the task sequences you want to prototype.
- Create system responses (pop-ups, progress bars...) that correspond to each action the user can do in the wireframes.
- Create dummy content (text, images, etc.) that would result from user actions (e.g., filling in the password).
- Try to ensure each user action transitions the prototype into a state that the prototype can support.
- Lo-Fi prototypes should support full task sequences.
- Transitions can often be effectively represented with very simple tools (e.g., cutouts, unfolding paper, etc.)

Make it 'Runnable'

- Result: a representation of the design that can be “run” to provide stakeholders with an experience of how the system would behave.

Summary

- Lo-fi prototypes are cheap, disposable ways to interactively test early design ideas.
- Low fidelity focuses users on design concepts and interaction rather than minor design details.
- Repeated prototyping and testing enables rapid iteration and refinement of the design.

Testing Lo-Fi Prototypes

- 1. Select Users
 - If feasible, test prototype with members of the target audience system is intended to support.
 - If target users difficult to find, identify their important skills/ context and find proxy users.
 - Avoid users who would not give you honest feedback.

Testing Lo-Fi Prototypes

- 2. Identify Testing Tasks

- Develop tasks that users should be able to accomplish using the prototype.
- Ensure the prototype supports these tasks fully.
- Create fake but realistic data to populate the prototype as user moves through tasks.
- Ensure that prototype can transition between tasks.

Testing Lo-Fi Prototypes

- 3. Find Testing Staff
 - Facilitator:
 - interacts with users and provides instructions.
 - Computer:
 - Changes state of the prototype in response to user actions.
 - Observer:
 - Takes notes about user reactions and their feedback.
- 4. Conduct Test
 - Ask users to complete each test task.
 - Provide only clarification feedback; allow users to try to figure out how to do the tasks.
 - Note where they get stuck, what confuses them, if they can't transition between states.
 - Discuss their experiences with using the prototype.

- 5. Evaluate and Use Test Results

- Identify common problems users experienced.
- Identify useful user suggestions for prototype improvement.
- Determine if any aspect of the prototype is sufficiently flawed that it needs to be completely rethought.
- Decide how to revise the prototype to account for feedback.
- Create new wireframes with revised functionality

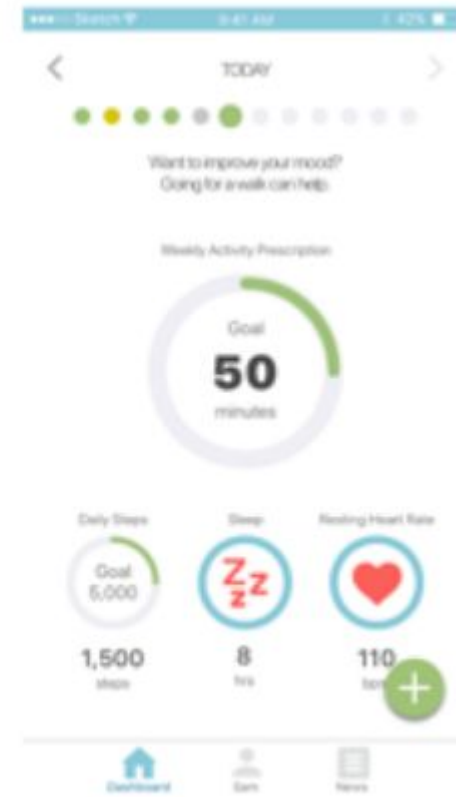
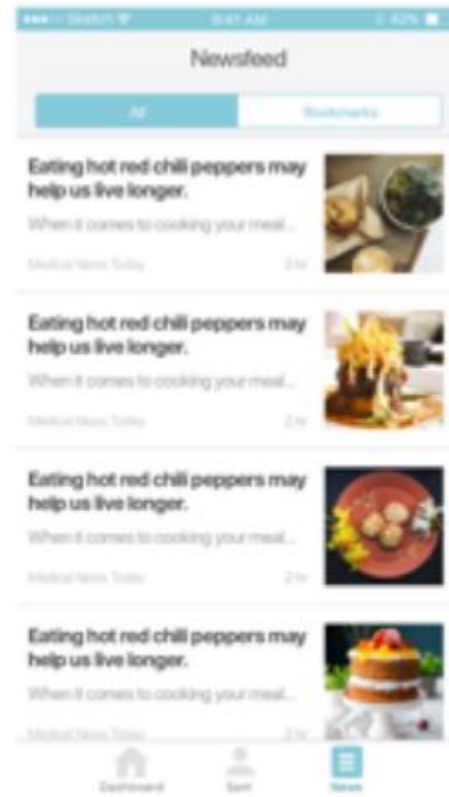
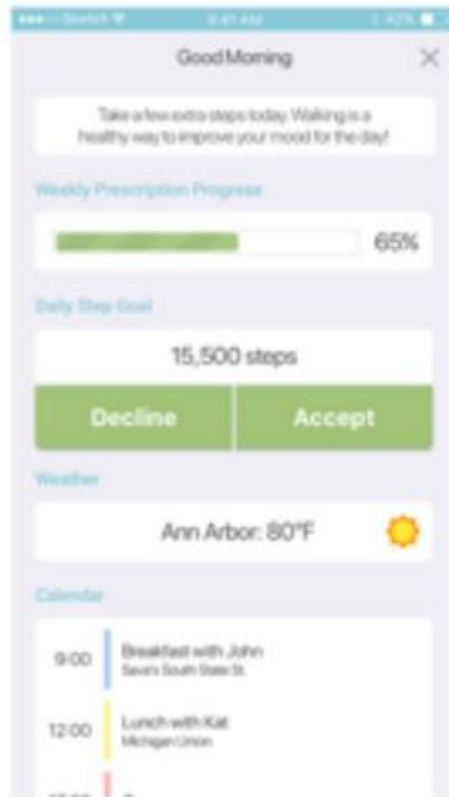
Tips

- Basic usability problems (e.g., navigation) often found after 1-2 testers.
- Revise the prototype before enrolling more testers.
- Often useful to give users time to just explore the prototype to familiarize themselves with it before starting test tasks.
- In post-test interview, get feedback both on the design concept and specific usability issues user encountered.
- Create and test a new prototype as soon as the design has sufficiently changed.

Classwork

- Search up “Wizard oz OZ” Prototypes

Mid-High Fidelity Prototypes



Hi-fi prototypes

- Prototypes that accurately reflect mature design and can be used as a part of the system specification to guide implementation.

Popular Prototyping Software

- Proto.io(<http://proto.io>)
- Invision(<http://invisionapp.com>)
- Marvel(<http://marvelapp.com>)
- JustinMind(<http://justinmind.com>)
- Framer (<http://framer.com>)
- Axure
- Sketch
- AdobeXD
- AdobeIllustrator

Summary

Type	Advantages	Disadvantages
Low-fidelity prototype	- Lower development cost - Evaluates multiple design concepts - Useful communication device - Addresses screen layout issues - Useful for identifying market requirements - Proof of concept	- Limited error checking - Poor detailed specification to code to - Facilitator-driven - Limited utility after requirements established - Limited usefulness for usability tests - Navigational and flow limitations
High-fidelity prototype	- Complete functionality - Fully interactive - User-driven - Clearly defines navigational scheme Use for exploration and test - Look and feel of final product Serves as a living specification - Marketing and sales tool	- More resource-intensive to develop - Time-consuming to create - Inefficient for proof-of-concept designs - Not effective for requirements gathering